Stage border bends — SOFC plant

How sharply the curves bend when the plant changes stage. Data 2025-10-30 to 2026-03-30, 1-minute, decided 8 plant stages. Made 2026-11-03.

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1 · Overview

The stack temperature bends gently on the normal start-up path. The heaters and gas flows bend much harder. The only sharp stack bends come when the plant goes into power (Fuel switch → Part load) and when it leaves power (trip or cool-down).

Why it matters: a ceramic stack cracks from fast changes in temperature, not from heat alone. These numbers show where the plant pushes the stack hardest, and give the realistic bend to copy in synthetic data and the twin.

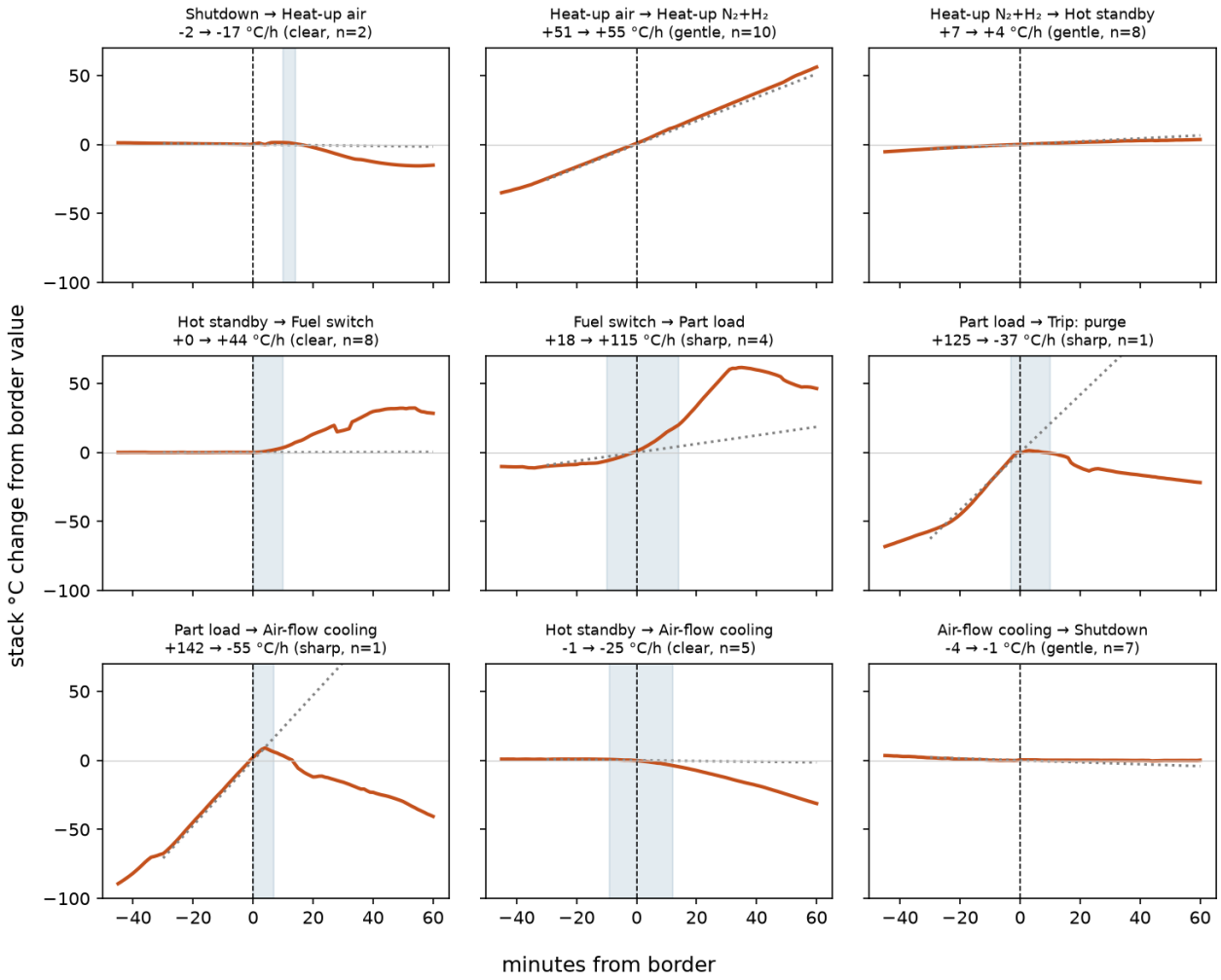
Three words. Think of driving. **Slope** is your speed (°C per hour). **Bend** is how much the speed changes at the border (after minus before). **Jerk** is how hard you stomp the pedal: the same bend done in 2 minutes is a bigger jerk than over 20 minutes.

Border	Runs	Before °C/h	After °C/h	Bend °C/h	Bend takes	Size
Shutdown → Heat-up air	2	-2	-17	-15	4 min	clear
Heat-up air → Heat-up N□+H□	10	+51	+55	+4	—	gentle
Heat-up N□+H□ → Hot standby	8	+7	+4	-2	—	gentle
Hot standby → Fuel switch	8	+0	+44	+43	10 min	clear
Fuel switch → Part load	4	+18	+115	+96	24 min	sharp
Part load → Trip: purge	1	+125	-37	-162	13 min	sharp
Part load → Air-flow cooling	1	+142	-55	-196	7 min	sharp
Hot standby → Air-flow cooling	5	-1	-25	-24	21 min	clear
Air-flow cooling → Shutdown	7	-4	-1	+4	—	gentle

Stack temperature, median of clean borders (each side ≥ 10 min, no logging gap). Gentle < 10, clear 10–50, sharp > 50 °C/h.

If you remember one thing: the stack bends gently on the normal start-up path; the hard bends are at entering and leaving power.

2 · The stack curve around each border

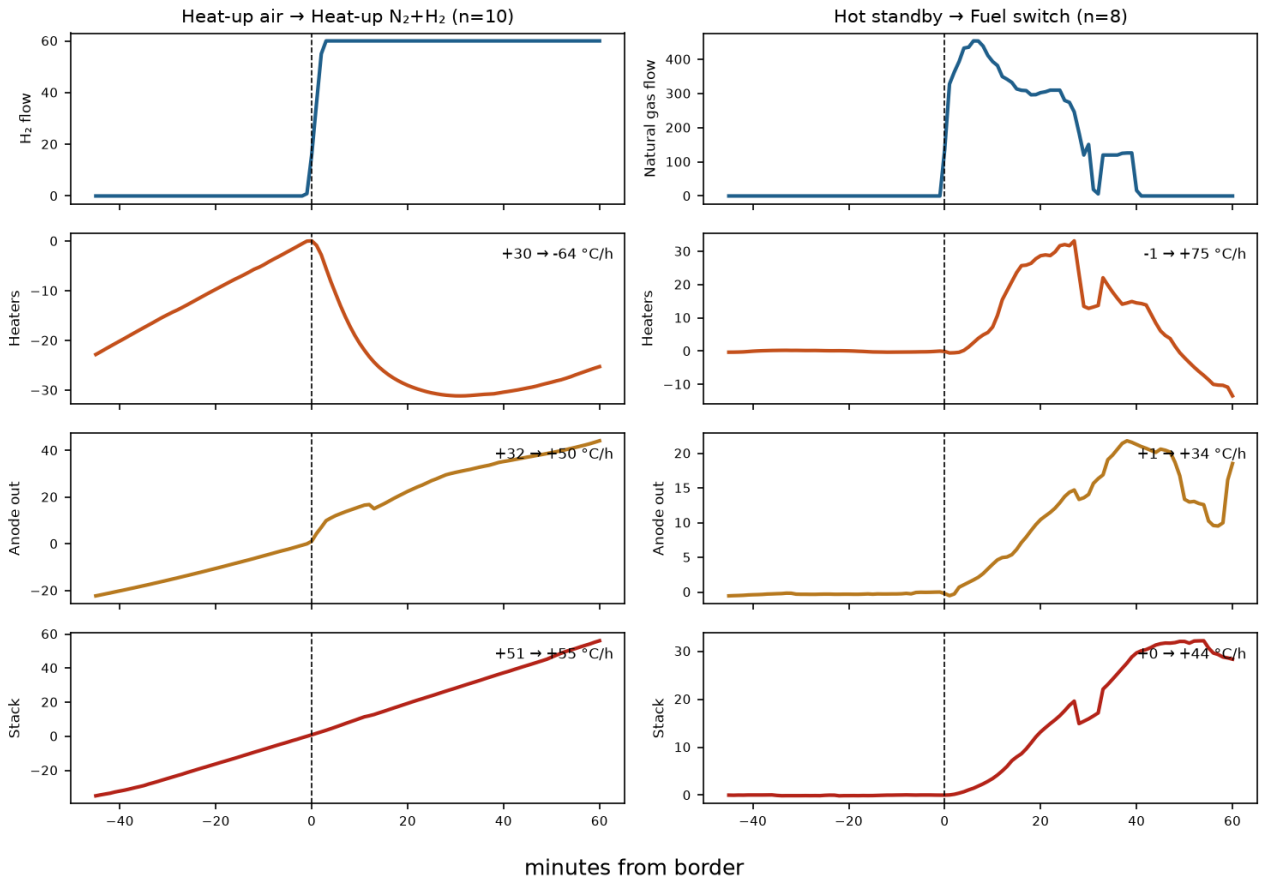


Orange = median stack change from the border value. Dotted = old slope carried on; the gap is the bend. Shaded = when the bend happens (10–90 %).

- Gas in does not jolt the stack: Heat-up air → Heat-up N□+H□ keeps ≈ +51…+55 °C/h.
- Going into power is the biggest normal bend: +19 → +115 °C/h over ≈ 24 min as current ramps.
- Leaving power is the sharpest: +125…+140 → −37…−55 °C/h in under 15 min (single runs).
- Starting heat-up first cools the stack (−17 °C/h for ≈ 15 min): cold air arrives before the heaters warm it.

If you remember one thing: a normal border bends the stack by less than 50 °C/h over 10–20 minutes; only entering or leaving power goes beyond that.

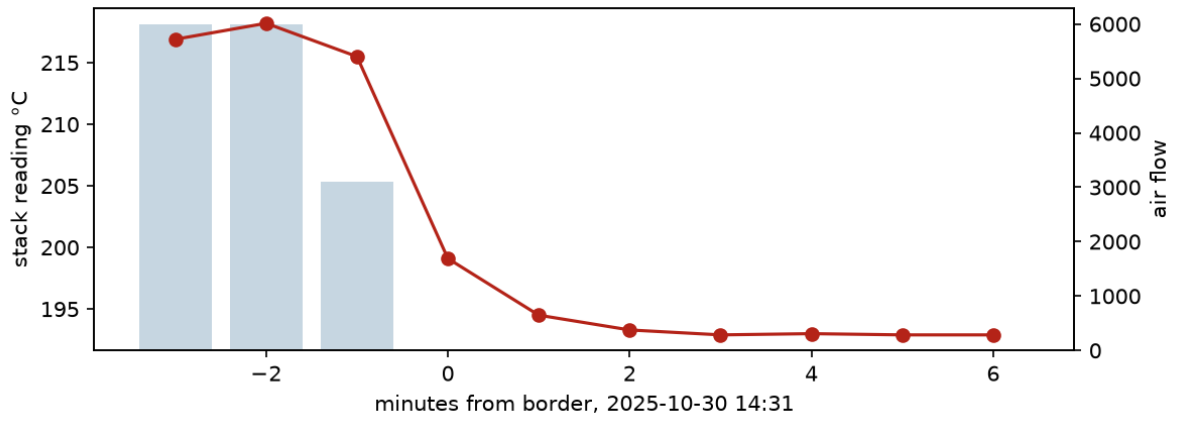
3 · Heaters and flows bend far harder than the stack



Flows are switched in one step (within one logged minute). The heaters react within minutes; the stack sits behind them and smooths the bend. At the gas switch the heaters swing from +31 to −64 °C/h while the stack hardly changes. At Hot standby → Fuel switch the heaters bend first (≈ 6 min), then anode out, then the stack (≈ 10 min). For flows the useful numbers are step size and follow-up time, not slope. Current ramps ≈ +11 A/h in Part load. Heaters often bend before the border because the command comes first.

If you remember one thing: flows jump, heaters jerk, the stack follows slowly. Judge stack stress on the stack curve.

4 · Odd cases, method, decisions



Fake bend when air stops. The stack reading drops ≈ 20 °C in two minutes and then stays flat (also 2026-03-17). Taken at face value it is −270 °C/h. It is a sensor settling when gas stops flowing past it, so it is left out of the bend figures and should be left out of any rate limit.

Leaving power has only one clean run per border type; use as examples.

Method. 131 borders found, 64 clean. Slope = straight-line fit 30 min each side. Bend time = 10 % → 90 % of the change in the 9-min smoothed slope. Acceleration (°C/h per min) and jerk (°C/h per min²) = peaks within ±15 min of the smoothed curve.

Border	Stack accel.	Stack jerk	Heaters accel.	Heaters jerk
Heat-up air → Heat-up N□+H□	1.6	0.3	30	7.3
Heat-up N□+H□ → Hot standby	0.5	0.2	0.4	0.1
Hot standby → Fuel switch	4.6	1.1	40	10.8
Fuel switch → Part load	9.1	1.3	43	12.6
Part load → Trip: purge	23	4.1	56	11.6
Part load → Air-flow cooling	30	5.2	27	4.9
Hot standby → Air-flow cooling	1.8	0.4	18	3.7
Air-flow cooling → Shutdown	6.1	2.2	7.9	2.8
Shutdown → Heat-up air	5.4	1.8	64	14.0

Decisions

1. Rate limit: do we have the stack maker’s maximum heating/cooling rate?
2. Generator: join stages with these bend times (≈ 10 min stack, ≈ 5 min heaters, flows as steps)?
3. Mark the 1–2 min after air stops as "sensor settling"?
4. Collect more Part load exits before using those bends in the twin?

If you remember one thing: the biggest stack "bend" in the data is a sensor effect when air stops; the biggest real ones are entering and leaving power.